Chapter 16. Logistic Regression

Logistic model, odds ratios, maximum likelihood and IRLS, 2×2 tables

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16.1 A model for a binary response

When y takes only the values 0 and 1, $E(y \mid x) = P(y = 1 \mid x) =: \pi(x)$ is a probability. The linear model $y = \beta_0 + \beta_1 x + \epsilon$ is unsuitable: fitted values can leave $[0, 1]$, the errors take only two values for each x (so they are not normal), and $\mathrm{Var}(y \mid x) = \pi(x)\{1 - \pi(x)\}$ varies with x. Empirically, $\pi(x)$ often follows an S-shaped curve. We therefore model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf G, the inverse **link function**. Common choices: the logistic $G(t) = e^t/(1 + e^t)$, the normal cdf Φ (probit), and the Gumbel $G(t) = \exp(-e^{-t})$. The logistic choice gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in x if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

Odds and odds ratios. The **odds** of $y = 1$ are $\pi/(1 - \pi) = P(y = 1)/P(y = 0) \in [0, \infty)$, with $\pi = \text{odds}/(1 + \text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1 - \pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1 - \pi(x+1)\}}{\pi(x)/\{1 - \pi(x)\}} = e^{\beta_1}.$$

So e^{β_1} is the **odds ratio** for a unit increase in x, the same at every x. For a binary x (exposed $= 1$), e^{β_1} is the odds ratio between the two groups; for instance $\hat\beta_1 = 2.0$ means the odds of $y = 1$ are $e^2 \approx 7.4$ times higher in the exposed group.

16.2 Estimation

Motivation. For a binary response the linear model fails (fitted probabilities outside $[0, 1]$, nonconstant variance). Logistic regression models the log-odds linearly, and its coefficients are estimated by maximum likelihood with iterative reweighted least squares.

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim$ Bernoulli(π_i) give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^{n} \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = \sum_{i=1}^{n} \{y_i \mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}.$$

Theorem 16.1

With $\pi(\beta) = (\pi_1, \ldots, \pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1 - \pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial \beta \partial \beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (if X has full column rank), so l is strictly concave and the MLE, when it exists, is the unique root of $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$.

Proof. $\frac{d}{dt}\log(1 + e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1 + e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1 - G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0 < \pi_i < 1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1 - \pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ∎

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y} - \pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on X with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta}\{\mathbf{z}^{(t)} - X\beta\}'W^{(t)}\{\mathbf{z}^{(t)} - X\beta\},$$

hence the name **iteratively reweighted least squares** (IRLS). A common start is $\beta^{(0)} = \mathbf{0}$. Since the Hessian does not depend on $\mathbf{y}$, Newton–Raphson coincides with Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates the asymptotic covariance of $\hat\beta$. Wald tests and intervals use $\hat\beta_j/\mathrm{se}(\hat\beta_j) \approx N(0,1)$; likelihood ratio tests compare $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ with a chi-square distribution.

If the two classes can be perfectly separated by a hyperplane in x, l increases without bound along a direction and the MLE does not exist.

16.3 Two-by-two tables

Motivation. With a single binary predictor, logistic regression reduces to a 2×2 table, and its slope is the log odds ratio. This connects the model with classical contingency-table analysis.

Binary exposure (x) and outcome (y) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts n_{xy}:

	$y=0$	$y=1$	total
$x=0$	n_{00}	n_{01}	n_{0+}
$x=1$	n_{10}	n_{11}	n_{1+}

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

Theorem 16.2

In the logistic model $\mathrm{logit}\, P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary x, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

Proof. The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of π_1, π_0 are the sample proportions n_{11}/n_{1+} and n_{01}/n_{0+}. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ∎

Why the odds ratio. In a case–control study the row proportions do not estimate π_x, because the numbers of cases and controls are fixed by design. But the odds ratio is symmetric in x and y:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

since both equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the joint cell probabilities. So the odds ratio, and hence the slope β_1 of a logistic regression, can be estimated from case–control data; only the intercept is distorted by the sampling.

Example 16.1

In a trial, 140 subjects receive a supplement and 140 a placebo; 17 and 31 catch a cold, respectively. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$: the odds of a cold are about halved, and $\hat\beta_1 = \log 0.49 = -0.72$. The standard error is $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, so the Wald statistic is -2.19, significant at the 5% level.

16.4 Selected exercises

Motivation. Three consequences of the likelihood equations: fitted and observed counts agree, the standard error used in Example 16.1 comes from the information matrix, and complete separation destroys the MLE.

Exercise 16.1 (Fitted counts equal observed counts)

(a) Show that the MLE of a logistic regression satisfies $X'\hat\pi = X'\mathbf{y}$; in particular, with an intercept, $\sum_i \hat\pi_i = \sum_i y_i$.
(b) For a binary x with all four cell counts n_{xy} positive, derive Theorem 16.2 directly from (a).

Solution [Supplement].

(a) By Theorem 16.1 the MLE solves $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$. The intercept row of X' is $\mathbf{1}'$, which gives $\sum \hat\pi_i = \sum y_i$: the expected number of successes under the fit equals the observed number. Each other regressor gives $\sum_i x_{ij}\hat\pi_i = \sum_i x_{ij} y_i$.

(b) With $\mathbf{x}_i = (1, x_i)'$ and $x_i \in \{0, 1\}$, $\hat\pi_i$ takes the value $\hat\pi_1$ when $x_i = 1$ and $\hat\pi_0$ when $x_i = 0$. The two equations of (a) read

$$n_{0+}\hat\pi_0 + n_{1+}\hat\pi_1 = n_{01} + n_{11}, \qquad n_{1+}\hat\pi_1 = n_{11}.$$

Hence $\hat\pi_1 = n_{11}/n_{1+}$ and $\hat\pi_0 = n_{01}/n_{0+}$, both in $(0, 1)$ because all cells are positive. Then $\hat\beta_0 = \operatorname{logit}\hat\pi_0$ and $\hat\beta_1 = \operatorname{logit}\hat\pi_1 - \operatorname{logit}\hat\pi_0$ solve the likelihood equations, and by the strict concavity of Theorem 16.1 they are the unique MLE. Finally $e^{\hat\beta_1} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ■

Exercise 16.2 (Standard error of the log odds ratio)

In the setting of Exercise 16.1(b), show that the entry of $\{X'W(\hat\beta)X\}^{-1}$ for β_1 equals

$$\frac{1}{n_{11}} + \frac{1}{n_{10}} + \frac{1}{n_{01}} + \frac{1}{n_{00}},$$

the squared standard error used in Example 16.1.

Solution [Supplement].

Step 1. $X'WX = \sum_i w_i \mathbf{x}_i \mathbf{x}_i'$ with $w_i = \hat\pi_i(1-\hat\pi_i)$. Grouping by x, with $a = n_{0+}\hat\pi_0(1-\hat\pi_0)$ and $b = n_{1+}\hat\pi_1(1-\hat\pi_1)$,

$$X'WX = a\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} + b\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} = \begin{pmatrix}a+b & b\\ b & b\end{pmatrix}.$$

Step 2. Its determinant is $(a+b)b - b^2 = ab > 0$, and the $(2,2)$ entry of the inverse is $(a+b)/(ab) = 1/a + 1/b$.

Step 3. By Exercise 16.1(b), $a = n_{0+}\cdot\frac{n_{01}}{n_{0+}}\cdot\frac{n_{00}}{n_{0+}} = \frac{n_{01}n_{00}}{n_{0+}}$, so $\frac{1}{a} = \frac{n_{00}+n_{01}}{n_{01}n_{00}} = \frac{1}{n_{01}} + \frac{1}{n_{00}}$. In the same way $\frac{1}{b} = \frac{1}{n_{11}} + \frac{1}{n_{10}}$. Adding gives the claim. ■

Exercise 16.3 (Complete separation)

Suppose there is a vector $\mathbf{b}$ with $\mathbf{x}_i'\mathbf{b} > 0$ whenever $y_i = 1$ and $\mathbf{x}_i'\mathbf{b} < 0$ whenever $y_i = 0$. Show that $l(\beta) < 0$ for all β but $l(t\mathbf{b}) \to 0$ as $t \to \infty$, so that the MLE does not exist.

Solution [Supplement].

Step 1. For every β, $0 < \pi_i < 1$, so each term $y_i \log\pi_i + (1-y_i)\log(1-\pi_i)$ is negative and $l(\beta) < 0$.

Step 2. Along $\beta = t\mathbf{b}$: if $y_i = 1$, the term is $\log\pi_i = -\log(1+e^{-t\mathbf{x}_i'\mathbf{b}}) \to 0$ since $\mathbf{x}_i'\mathbf{b} > 0$; if $y_i = 0$, it is $\log(1-\pi_i) = -\log(1+e^{t\mathbf{x}_i'\mathbf{b}}) \to 0$ since $\mathbf{x}_i'\mathbf{b} < 0$. Hence $l(t\mathbf{b}) \to 0$.

Step 3. So $\sup_\beta l(\beta) = 0$, but no β attains it by Step 1: the MLE does not exist. Along the direction $\mathbf{b}$ the fitted probabilities tend to 0 or 1; in practice separation shows up as huge coefficients and fitted probabilities numerically equal to 0 or 1. ■